

Question		Answer	Mark	Mark scheme	Additional guidance
12	(a)	$\frac{4x - 6}{3x - 9}$	M1	factorises numerator of $4x^2 - 9$ eg $(2x - 3)(2x + 3)$ oe	$\frac{2x(2x - 3)(2x + 3)}{3x(2x + 3)(x - 3)}$
			M1	factorises denominator eg $x(x-3)$ or $3(2x + 3)$ or for $3x(2x^2 - 3x - 9)$	
			A1	cancel to give $\frac{4x - 6}{3x - 9}$	Accept $a = 4, b = -6, c = 3, d = -9$
	(b)	$\frac{-x + 8}{x(x + 1)(x - 2)}$	M1	method to use a common denominator eg $x(x+1)(x-2)$ by multiplying terms	Method must involve finding equivalents for all three separate terms; may be done in several stages.
			M1	deduce numerator eg $3x(x-2) + x(x+1) - 4(x+1)(x-2)$	
			A1	oe	Equivalents must be algebraically equivalent and must have involved full simplification.